

KayCare HMS — Enterprise Architecture & Comprehensive Client Overview

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Target Architecture: Multi-Tenant Healthcare Enterprise Suite

Technical Standards: ASP.NET Core 8 | React 18 | PostgreSQL | EF Core 8 | HL7 MLLP

1. Executive Summary & Unified Synergy

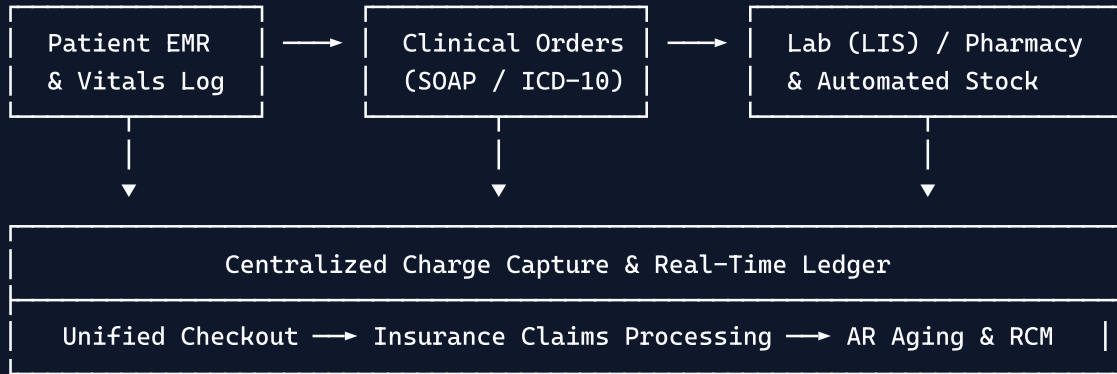
KayCare Hospital Management System (HMS) is an enterprise-grade, multi-tenant digital health platform engineered to unify complex clinical workflows, inpatient care, pharmacy operations, diagnostic laboratories, and financial revenue cycles into a single high-performance system.

The Unified Care Advantage

Traditional healthcare facilities struggle with fragmented, siloed software applications operating independently across departments. When Electronic Medical Records (EMR), Laboratory Information Systems (LIS), Pharmacy Management, and Billing operate in isolation, organizations experience severe operational friction:

Problem Domain	Legacy Siloed Systems	The KayCare HMS Unified Solution
Data Synchronization	Manual transcription of orders and lab results between isolated screens leads to clinical errors.	Instant Cross-Module Routing: Doctor orders flow dynamically to lab worklists and pharmacy dispensing queues.
Financial Integrity	Diagnostic tests and dispensed medications frequently bypass billing due to disconnected ledgers.	Automated Charge Capture: Every clinical event instantly registers items onto a centralized patient invoice.
Stock & Dispensing	Pharmacy inventory levels operate independently from clinical prescription entries.	Real-Time Stock Deductions: Dispensing drugs instantly adjusts active inventory balances and flags reorder alerts.
Multi-Facility Isolation	Cross-tenant data leaks and complex custom logic required for multi-clinic management.	Architectural Multi-Tenancy: EF Core Global Query Filters enforce absolute data isolation per clinic at the database level.

KAYCARE HMS UNIFIED ENTERPRISE ENGINE



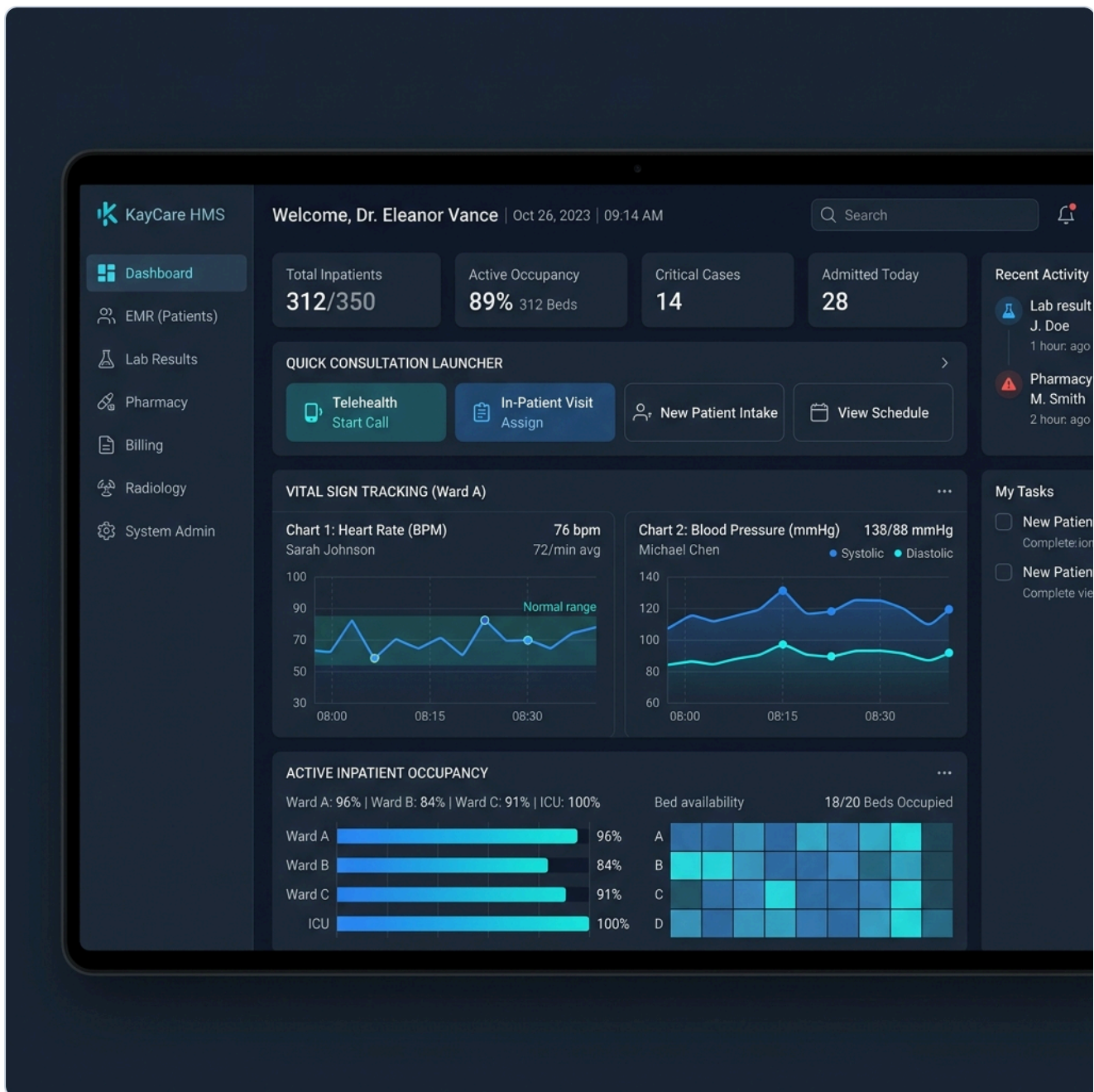
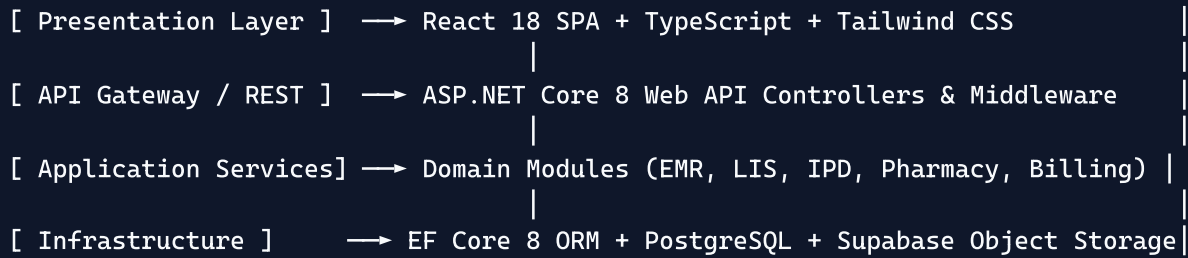


Figure 1.1: KayCare HMS Executive Clinical Dashboard featuring real-time bed occupancy, vitals tracking, and multi-department navigation.

2. System Architecture & Core Modules

KayCare HMS is built as a **Modular Monolith**. This architectural pattern combines the simplified deployment and strict transactional consistency of a monolithic application with the clean domain boundaries and decoupled design of microservices.

KAYCARE CORE ARCHITECTURE



2.1 Patient EMR & Outpatient Consultations

The Electronic Medical Record (EMR) serves as the digital source of truth for patient history and clinical notes.

- **Centralized Patient Registry:** Auto-generates unique, standardized Medical Record Numbers (MRNs). Tracks demographic profiles, emergency contacts, and insurance attachments.
- **Vitals & Allergy Tracking:** Logs longitudinal vital signs (Blood Pressure, Heart Rate, Respiratory Rate, O2 Saturation, Temperature, BMI) with visual progression graphs. Supports severe allergy flagging with automated clinical alerts during prescription writing.
- **Outpatient SOAP Documentation:** Facilitates structured clinical notes structured by **S**ubjective history, **O**bjective findings, **A**ssessment diagnoses, and **P**lan interventions.
- **ICD-10 Codification Engine:** Integrated clinical coding search allowing physicians to map primary and secondary diagnoses to standard ICD-10 codes.

2.2 Inpatient Department (IPD) & Nursing Workspace

Designed to handle complete hospital admission cycles from bed booking to discharge summary generation.

- **Dynamic Ward & Bed Management:** Visually tracks ward categories (General Ward, Semi-Private, Private VIP, ICU, HDU) and bed occupancy states (Available, Occupied, Maintenance, Cleaning).
- **Admission & Transfer Workflows:** Streamlines inpatient admission, attending physician assignment, ward transfers, and automated daily room rate charge capture.
- **Medication Administration Record (MAR):** Interactive nursing log interface for verifying dosage timing, administrator ID, and administration status (Administered, Refused, Held).
- **Nursing Care Notes & Flowsheets:** Shift notes and vital sign monitoring sheets synchronized across nursing stations.

2.3 Laboratory Information System (LIS) & Diagnostic Engine

The LIS module automates diagnostic workflows from initial physician order to final validated lab report.

- **Sample Accessioning & Lifecycle:** Tracks test specimens through explicit status transitions: **Ordered** → **Sample Received** → **Resulted** → **Signed Off**.
- **HL7 MLLP Machine Connectivity:** Features out-of-the-box support for Health Level 7 (HL7) Minimum Lower Layer Protocol (MLLP) messaging. Ingests automated test results directly from laboratory analyzer equipment without manual data entry.
- **Critical Result Alerts:** Highlights panic-level laboratory results in real-time, enforcing mandatory logged call entries to confirm direct verbal notification to the ordering physician.

KayCare LIS

Search

Dr. Sarah Chen

LIS Admin

Dashboard

Accession

Results

Reports

Settings

Connected

Print

LIS Workspace

Accession Tracking

Save

Release

Print

Accession # ↑	Patient Name	DOB/Sex	Received	Status	HL7 Device
LAB-98765	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Ordered	● Sysmex XN-10
LAB-98766	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Sample Received	● Roche c501
LAB-98763	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Ordered	● Roche c501
LAB-98764	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Resulted	● Sysmex XN-10
LAB-98765	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Resulted	● Roche c501
LAB-98766	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Signed Off	● Resulee
LAB-98767	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Resulted	● Resulee
LAB-98768	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Signed Off	● Roche c501
LAB-98769	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Resulted	● Resulee
LAB-98760	John Doe (45M)	2023-10-26 09:30	2023-10-26 10:15	Signed Off	● Resulee

10 recents

Result Entry & Review

Accession # LAB-98765 | Jane Smith (22F) | Received: 2023-10-26 10:15 | Status: Resulted

Hematology (CBC)

Resulted

Test	Result	Ref. Range	Flag
WBC	14.5	52.0 - 100.0	
RBC	14.5	1.45 - 1.25	H
Hgb	9.2	9.2 - 100	L
Hct	13.5	18.5 - 25%	
Plt	45k	45k - 45k	LL

Biochemistry

Test	Result	Ref. Range	Flag
Boentology	Resulted	Resulted	H/L
WBC	100	120 - 100	H
Hgb	14.5	125 - 190	H
Hct			
Plt	45k	45k - 3k	LL

Figure 2.1: KayCare HMS Diagnostic Laboratory Information System featuring sample accessioning, HL7 analyzer links, and critical value flags.

2.4 Pharmacy, Drug Formulary & Inventory Lifecycle

Provides end-to-end control over pharmaceutical stock, prescription fulfillment, and supplier purchasing.

- **Prescription Workspace:** Direct electronic prescribing from clinical consultations. Enforces dosage validation, refill limits, and controlled drug safety checks.
- **Real-Time Inventory Deductions:** Automatically calculates and subtracts exact unit quantities from drug stock balances upon counter dispensing.
- **Controlled Substances Register (CS Register):** Dedicated, audited logging register for schedule II-V medications to meet regulatory compliance standards.
- **Inventory & Purchasing Lifecycle:** Manages supplier catalogues, generates Purchase Orders (POs), logs Goods Received Notes (GRN), and triggers automated low-stock reorder alerts.

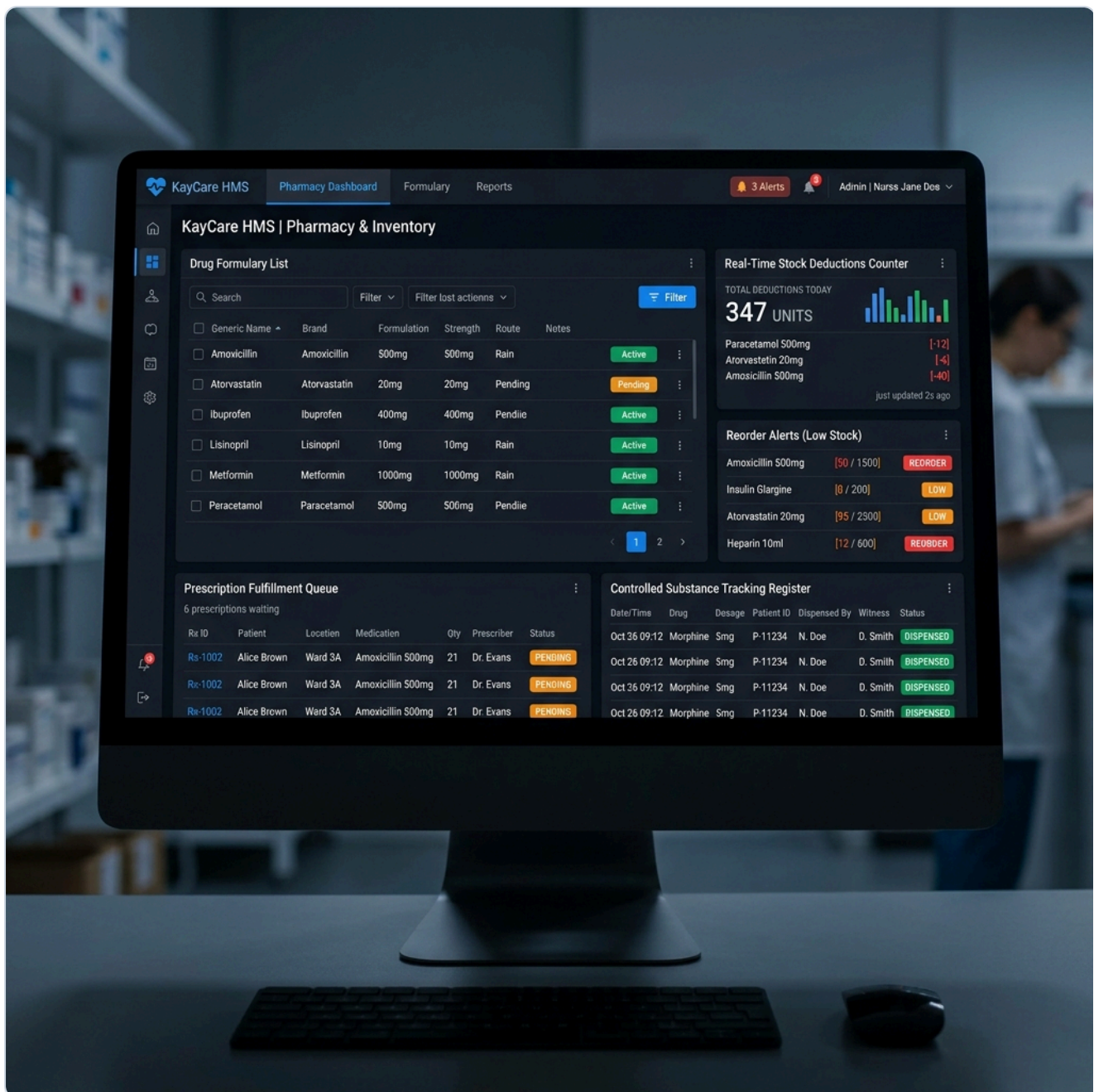


Figure 2.2: KayCare HMS Pharmacy & Drug Inventory Workspace showing real-time stock balances, dispensing queues, and reorder alerts.

2.5 Radiology & PACS Imaging

- **Procedure Order Dispatch:** Schedules radiology exams (X-Ray, Ultrasound, CT, MRI) and manages radiology technician worklists.
- **Integrated DICOM Viewer:** Embedded zero-footprint WebGL DICOM viewport allowing medical staff to view high-resolution diagnostic images directly in the browser without installing external client software.

2.6 Unified Billing Ledger & Revenue Cycle Management (RCM)

- **Centralized Charge Capture Engine:** Consolidates consultation fees, nursing services, daily ward bed rates, laboratory tests, and dispensed drugs onto a single master patient bill.
- **Financial Ledger Flexibility:** Supports line-item adjustments, authorized discounts, partial payments, credit notes, and multi-method refund processing.
- **Insurance Claims Engine:** Converts patient invoices into formatted claims for submission to private health insurance companies and national health insurance schemes (e.g., NHIS).
- **Accounts Receivable (AR) Aging:** Provides real-time aging reports (0-30 days, 31-60 days, 61-90 days, 90+ days) and financial performance dashboards.



Figure 2.3: KayCare HMS Unified Billing Ledger featuring itemized revenue breakdowns, insurance claim trackers, and AR aging analysis.

3. Technology Stack & Deployment Architecture

KayCare HMS leverages modern open enterprise technology to ensure maximum performance, developer productivity, and hosting cost efficiency.

TECHNOLOGY STACK		
Layer	Technology / Framework	Strategic Purpose
Frontend App	React 18, Vite, TypeScript Tailwind CSS	Rapid render SPA, strict type safety & modern UI.
Backend Web API	ASP.NET Core 8 Web API Entity Framework Core 8	High-throughput C# engine, asynchronous I/O pipelines.
Database Engine	PostgreSQL Relational DB JSONB support & indexes	Enterprise ACID integrity, robust concurrency controls
Object Storage	Supabase Storage / S3 Secure Signed URLs	Secure, scalable storage for DICOM & PDF documents.

Infrastructure & Cloud Delivery

- **Backend Containerization:** The ASP.NET Core backend is compiled into lightweight Docker images. It can be hosted on modern container platforms such as **Render**, **Railway**, **AWS ECS**, or **Azure App Service**.
- **Global Edge Content Delivery:** The React SPA frontend build is optimized for edge hosting on global Content Delivery Networks (CDNs) like **Vercel**, **Netlify**, or **Cloudflare Pages**, providing sub-50ms loading times globally.

4. Security, Compliance, & Multi-Tenant Reliability

Healthcare data demands the highest standard of defensive application design and security architecture.

4.1 Multi-Tenant Data Isolation Strategy

KayCare HMS enforces strict, bulletproof tenant isolation at the database access layer.

- **EF Core Global Query Filters:** Every database query automatically appends a `TenantId` predicate (`WHERE TenantId = @CurrentTenant`).
- **Tenant Resolution Middleware:** Incoming HTTP requests are intercepted by `TenantResolutionMiddleware` , which validates the `X-Tenant-Code` header or user claim, establishing the security context before any business logic executes. This prevents cross-tenant data leaks.

4.2 HIPAA Audit Logging & Non-Repudiation

To satisfy HIPAA and ISO 27001 regulatory compliance, KayCare HMS implements an immutable audit ledger service (`AuditLog`).

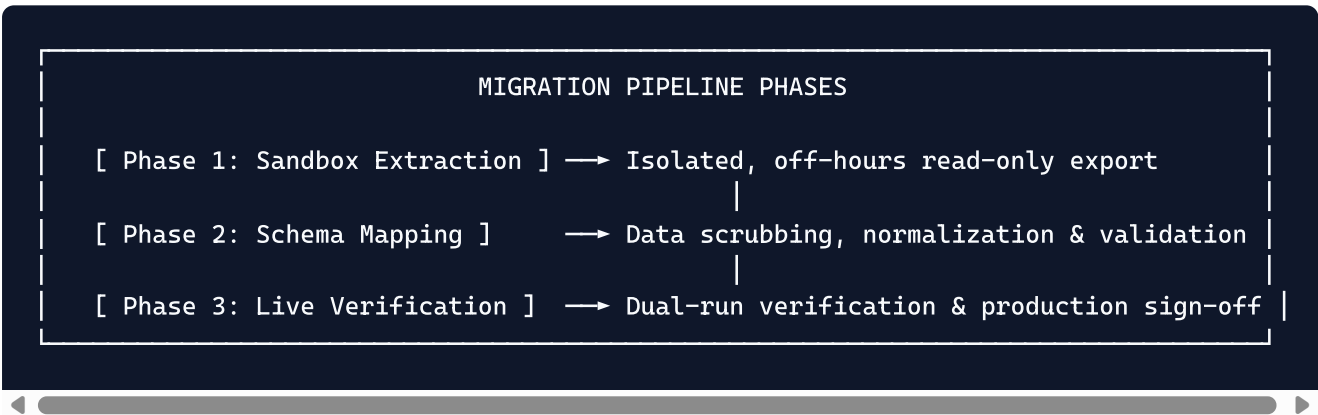
- **Granular Tracking:** Captures Staff ID, Patient MRN, Action Type (Create, Update, Delete, View), Exact Endpoint, Client IP Address, and Timestamp.
- **State Delta Storage:** Logs JSON representations of `OldValues` and `NewValues` for financial and medical record changes, ensuring historical auditability and non-repudiation.

4.3 Transaction Atomicity & Concurrency Control

- **ACID Financial Transactions:** All billing checkouts, claim payouts, and inventory stock deductions run inside explicit database transactions. If any sub-step fails or loses connection, the entire transaction rolls back instantly to eliminate discrepancies.
- **Optimistic & Pessimistic Locking:** Prevents race conditions during simultaneous drug stock dispensing or bed bookings across multiple nursing stations.

5. Risk-Free Data Migration Protocol

Transitioning from legacy hospital management systems to KayCare HMS is achieved with zero operational downtime using a structured three-phase pipeline.



1. Phase 1: Secure Extraction & Auditing

Legacy databases (Demographics, historical consultation records, item catalogs, outstanding

financial balances) are extracted into an encrypted staging environment. Extraction scripts run off-peak to avoid affecting clinical operations.

2. **Phase 2: Schema Mapping & Normalization**

Legacy records are programmatically scrubbed, deduplicated, and transformed into KayCare HMS PostgreSQL relational models. Automated integrity constraints verify that all clinical notes reference valid patient MRNs before live loading.

3. **Phase 3: Parallel Validation & Cutover Sign-Off**

A validation layer compares historical totals and ledger balances between legacy tools and KayCare HMS. Once verified, production deployment is finalized, providing clinical staff with complete patient medical timelines from Day One.